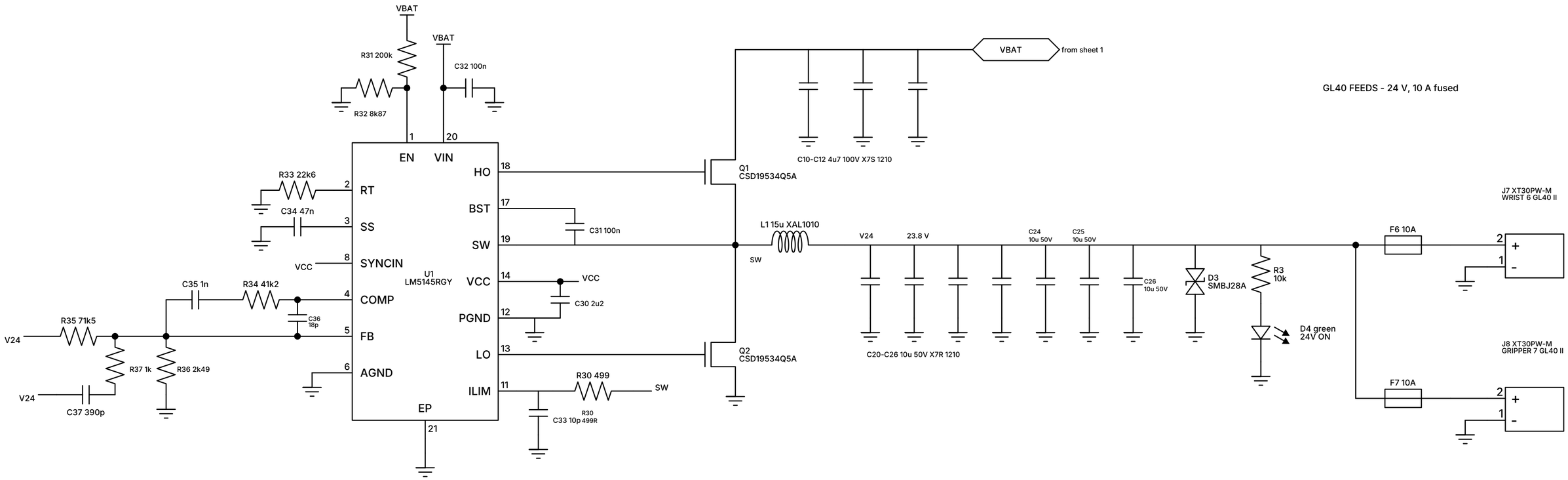
24 V BUCK - TI LM5145 (SNVSAI4B, application circuit 3: 48 V -> 24 V, 440 kHz, 5 A cont.)




NOTES

1. Values exactly as TI's application circuit 3 (compensation is TI's); RFB1 71k5 (E96) sets 23.8 V.

2. SYNCIN tied to VCC: forced PWM, so GL40 regen flows back to VBAT instead of raising V24.

3. 5 A continuous; the valley limit (R30 on Q2's RDS(on)) lets ~8-9 A out for peaks.

4. UVLO: R31 / R32 start the converter at 28.3 V, stop it at 26.3 V.

5. J7 / J8 are XT30 male: a GL40 lead cannot reach a 49 V socket, nor an AK lead a 24 V one.

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| 24 V CONVERTER + GL40 FEEDS | |
| PIONEER-ARM-PDB (cad_bench/projects/arm_pdb.py) | |
| TI LM5145 application circuit 3, unchanged | |
| SHEET 2 / 2 | REV A |
| DATE 2026-10-12 | SIZE A3 |